

Vector-Based Intelligent Medical Data Lake Using Cosine Similarity Clustering

Abstract

This report presents a healthcare data platform that uses vector embeddings and cosine similarity to organize and retrieve medical documents semantically.

Problem Statement

Hospitals generate millions of reports. Traditional folder structures and keyword search make retrieval difficult. The proposed system stores embeddings in a vector database and groups similar reports into clusters.

Objectives

Build a scalable platform for ingestion, embedding generation, similarity search, clustering, secure storage, and semantic retrieval.

Architecture

Data Ingestion → OCR/Text Extraction → Embedding Generation → Cosine Similarity → Vector Database → Cluster Assignment → Semantic Search → Dashboard.

Methodology

Extract text from uploaded reports, generate embeddings using a transformer model, compare against existing vectors with cosine similarity, assign to an existing cluster or create a new one, and store metadata plus vector index.

Cosine Similarity

$\cos(\theta) = \frac{A \cdot B}{|A||B|}$. Higher values indicate semantically similar reports.

Technology Stack

Python, FastAPI, PostgreSQL, Qdrant/Milvus, Sentence Transformers, Docker, Streamlit/React.

Team Roles

1) Data ingestion 2) OCR & preprocessing 3) Embeddings 4) Similarity & clustering 5) Backend & vector DB 6) Dashboard & deployment.

Advantages

Fast semantic retrieval, scalable organization, reduced duplication, improved clinical search, secure metadata management.

Future Scope

RAG integration, multimodal medical imaging embeddings, federated hospital search, AI-assisted diagnosis.